

Standard Tolerances Guide

General machining tolerances are controlled by customer drawing, approved QAP (Quality Assurance Plan), and applicable ISO / DIN / ASME standards.

Typical Achievable Tolerances

Turned Diameters	Down to ± 0.005 mm where geometry permits
Ground Bearing Fits	Ra 0.2 to 0.8 microns
Face Runout	0.01 mm to 0.02 mm depending on diameter
Shaft Straightness	0.05 mm per metre
Concentricity	Within 0.01 mm
Surface Finish	10 to 50 microns
Keyways & Slots	Inspected by calibrated gauges or CMM
Bore Tolerance	H7 / h6 standard fits or custom
Thread Profiles	Verified with calibrated thread gauges

Inspection Equipment

- Coordinate Measuring Machine (CMM)
- Calibrated micrometers and bore gauges
- Surface roughness tester
- Dial indicators and V-blocks
- Thread plug and ring gauges
- Digital height gauges

Note

All dimensions remain subject to drawing review and manufacturability confirmation. Contact us with your specific drawing for a detailed tolerance assessment.

Contact

Email sales@pakoshaft.com for drawing-based tolerance quotations.

